

Bulan 8: Advanced Specializations & Portfolio

Minggu 1: IoT & Real-Time Systems

Hari 1: IoT Device Management

Project: DeviceHub

- Deskripsi: IoT device monitoring dan control platform
- Fitur: Device registration, telemetry, commands, firmware updates
- Keunikan: MQTT protocol, device shadows, edge computing

Hari 2: Smart Home Automation

Project: HomeAI

- Deskripsi: Smart home automation dengan rule engine
- Fitur: Device control, scenes, automation rules, voice commands
- Keunikan: ML-based behavior learning, energy optimization

Hari 3: Fleet Management System

Project: FleetTrack

- Deskripsi: Vehicle tracking dan logistics management
- Fitur: GPS tracking, route optimization, maintenance, fuel monitoring
- Keunikan: Geofencing, driver behavior analytics

Hari 4: Environmental Monitoring

Project: EcoWatch

- Deskripsi: Environmental sensor network platform
- Fitur: Air quality, temperature, humidity monitoring, alerts
- Keunikan: Predictive analytics, crowd-sourced data

Hari 5: Industrial Automation Dashboard

Project: FactoryView

- Deskripsi: Manufacturing monitoring dengan OEE tracking

- Fitur: Machine status, production metrics, downtime analysis
- Keunikan: Predictive maintenance, anomaly detection

Hari 6: Energy Management System

Project: PowerOptimize

- Deskripsi: Energy consumption tracking dan optimization
- Fitur: Usage monitoring, cost analysis, device scheduling
- Keunikan: Solar panel integration, grid optimization

Hari 7: Smart Agriculture Platform

Project: FarmTech

- Deskripsi: Precision agriculture dengan sensor integration
- Fitur: Soil monitoring, irrigation control, crop health, weather
- Keunikan: AI crop disease detection, yield prediction

Minggu 2: Media & Content Platforms

Hari 8: Video Streaming Platform

Project: StreamHub

- Deskripsi: Video hosting dengan adaptive streaming
- Fitur: Upload, transcoding, HLS/DASH, subtitles, DRM
- Keunikan: AI content moderation, thumbnail generation

Hari 9: Podcast Platform

Project: PodFlow

- Deskripsi: Podcast hosting dan distribution
- Fitur: RSS feeds, analytics, monetization, transcription
- Keunikan: Auto-chapter detection, episode recommendations

Hari 10: Digital Asset Management

Project: MediaVault

- Deskripsi: Enterprise DAM system

- Fitur: Asset library, metadata, permissions, CDN integration
- Keunikan: AI tagging, brand compliance checker

Hari 11: News Aggregator

Project: NewsFlow

- Deskripsi: Personalized news aggregation platform
- Fitur: RSS parsing, categorization, personalization, bookmarks
- Keunikan: Fake news detection, reading time estimation

Hari 12: Music Streaming Service

Project: SoundWave

- Deskripsi: Music streaming dengan playlist curation
- Fitur: Library, playlists, radio, lyrics, offline mode
- Keunikan: AI DJ mode, mood-based recommendations

Hari 13: Photo Sharing Platform

Project: SnapShare

- Deskripsi: Photo social network (Instagram-like)
- Fitur: Feed, stories, filters, hashtags, explore
- Keunikan: AI photo enhancement, collaborative albums

Hari 14: Content Creator Platform

Project: CreatorSpace

- Deskripsi: Platform untuk content creators (Patreon-like)
- Fitur: Subscriptions, exclusive content, tips, analytics
- Keunikan: Tokenized rewards, creator tools suite

Minggu 3: Social & Community

Hari 15: Social Network Platform

Project: SocialSphere

- Deskripsi: Full-featured social network
- Fitur: Profiles, posts, comments, likes, friends, messages, groups
- Keunikan: Privacy controls, content moderation AI

Hari 16: Forum & Community Platform

Project: ForumPro

- Deskripsi: Discussion forum dengan gamification
- Fitur: Topics, threads, replies, reputation, badges
- Keunikan: Topic clustering, expert identification

Hari 17: Dating App Backend

Project: MatchMaker

- Deskripsi: Dating app dengan matching algorithm
- Fitur: Profiles, swiping, matches, chat, video calls
- Keunikan: Compatibility scoring, safety features

Hari 18: Event & Meetup Platform

Project: GatherHub

- Deskripsi: Local events dan meetup organizer
- Fitur: Event creation, RSVPs, check-ins, networking
- Keunikan: Interest-based suggestions, attendee matching

Hari 19: Q&A Platform

Project: AskExperts

- Deskripsi: Stack Overflow-style Q&A
- Fitur: Questions, answers, voting, reputation, tags
- Keunikan: AI answer validation, bounty system

Hari 20: Volunteer Management

Project: GoodDeeds

- Deskripsi: Volunteer coordination platform

- Fitur: Opportunity listings, applications, scheduling, hours tracking
- Keunikan: Impact metrics, skill-based matching

Hari 21: Alumni Network

Project: AlumniConnect

- Deskripsi: University alumni networking platform
- Fitur: Directory, job board, events, fundraising, mentorship
- Keunikan: Career path visualization, batch reunions

Minggu 4: Portfolio & Career Preparation

Hari 22: Portfolio Website Builder

Project: DevFolio

- Deskripsi: Developer portfolio generator dengan CMS
- Fitur: Project showcase, blog, resume, contact, analytics
- Keunikan: GitHub integration, one-click deploy

Hari 23: Code Snippet Manager

Project: SnippetBox

- Deskripsi: Personal code snippet library dengan sharing
- Fitur: Syntax highlighting, tags, search, private/public
- Keunikan: VS Code extension, CLI tool

Hari 24: Technical Blog Platform

Project: DevBlog

- Deskripsi: Developer blogging dengan markdown
- Fitur: Posts, series, tags, code blocks, comments
- Keunikan: SEO optimization, reading stats, newsletter

Hari 25: Open Source Project Hub

Project: OpenCollab

- Deskripsi: Platform untuk discover open source projects

- Fitur: Project listings, contribution tracking, mentorship
- Keunikan: Good first issue finder, skill matcher

Hari 26: Interview Preparation Platform

Project: CodePrep

- Deskripsi: Coding interview practice platform
- Fitur: Problem sets, mock interviews, solution discussions
- Keunikan: Company-specific tracks, peer interviews

Hari 27: Freelance Marketplace

Project: FreelanceHub

- Deskripsi: Freelancer dan client matching platform
- Fitur: Job posts, proposals, contracts, milestones, escrow
- Keunikan: Skill verification, dispute resolution

Hari 28: Documentation Generator

Project: DocsGen

- Deskripsi: Automatic documentation dari Django project
- Fitur: API docs, model diagrams, usage examples
- Keunikan: Interactive API explorer, changelog automation

Hari 29-30: Final Capstone Project

Project: TalentMarket

- Deskripsi: Comprehensive job marketplace platform
- Fitur:
 - Job seeker: Profile, resume builder, job search, applications, interview scheduler
 - Employer: Company profiles, job postings, applicant tracking, video interviews
 - Platform: Skill assessments, AI matching, salary insights, referral program
 - Advanced: Chat system, notification engine, analytics dashboard, mobile API
- Keunikan:
 - AI-powered resume screening

- Video interview analysis (sentiment, keywords)
 - Blockchain-verified credentials
 - Gig economy integration
 - Career path recommendations
 - Tech Stack Integration:
 - Django + DRF for backend
 - GraphQL for flexible queries
 - Celery for background jobs
 - Redis for caching
 - PostgreSQL with full-text search
 - WebSockets for real-time features
 - Docker containerization
 - CI/CD pipeline
 - Comprehensive testing
 - Scope: 2 hari untuk production-ready job platform
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Selamat! Kurikulum 8 Bulan Selesai

Skills Yang dikuasai:

-  Python Advanced & OOP
-  Django Core & Advanced Features
-  REST & GraphQL APIs
-  Real-time dengan WebSockets & Channels
-  Database Optimization & Scaling
-  Security Best Practices
-  Testing & CI/CD
-  Docker & Kubernetes
-  ML/AI Integration
-  30+ Production-Ready Projects

Portfolio Projects (Recommended untuk GitHub):

1. CommunityHub (Bulan 2)

2. MarketPlace (Bulan 3)
3. EventMaster (Bulan 4)
4. SaaSstarter (Bulan 5)
5. LearningPlatform (Bulan 6)
6. ERPLite (Bulan 7)
7. TalentMarket (Bulan 8)

Next Steps:

- Deploy projects ke production (AWS/GCP/Heroku)
- Contribute ke open source Django projects
- Build dan launch startup idea
- Apply for Django developer positions
- Mentor others in community

Selamat berkarir sebagai Django Developer! 🚀